require a major R&D effort on nuclear rockets, but it is still three hundred times too slow for *Interstellar*'s needs.

Fortunately, Nature provides a way to achieve the huge speed changes, *c*/3, required in *Interstellar:* gravitational slingshots around black holes far smaller than Gargantua.

Slingshot Navigation to Miller's Planet

Stars and small black holes congregate around gigantic black holes like Gargantua (more on this in the next section). In my science interpretation of the movie, I imagine that Cooper and his team make a survey of all the small black holes orbiting Gargantua. They identify one that is well positioned to gravitationally deflect the Ranger from its near circular orbit and send it plunging downward toward Miller's planet (Figure 7.2). This gravity-assisted maneuver is called a "gravitational slingshot," and has often been used by NASA in the solar system—though with the gravity coming from planets rather than a black hole (see the end of the chapter).

This slingshot maneuver is not seen or discussed in *Interstellar*, but the next one *is* mentioned, by Cooper: "Look, I can swing around that *neutron star* to decelerate," he says. Deceleration is necessary because, having fallen under Gargantua's huge gravitational pull, from the *Endurance*'s orbit to Miller's orbit, the Ranger has acquired too much speed; it is moving *c*/4 faster than Miller's planet. In Figure 7.3, the neutron star, traveling leftward relative to Miller's planet, deflects and slows the Ranger's motion so it can rendezvous gently with the planet.